

Darukaa.Earth: AI Biodiversity Intelligence Chatbot Challenge

Welcome to the Darukaa.Earth Hackathon. This challenge is designed to evaluate your ability to build **intelligent, knowledge-driven AI systems** that can reason about real-world environmental problems.

We are not looking for UI-heavy applications. We are looking for **depth of thinking, system design, and scientific reasoning**.

Objective

Build an AI-powered conversational system that:

- Maintains a **structured knowledge base** of biodiversity and environmental metrics
- Understands user queries about ecosystems, land, and climate conditions
- Generates **actionable, non-obvious recommendations** to improve biodiversity
- Supports every recommendation with **scientific reasoning and evidence**

Core Requirements

1. Knowledge System (Critical)

Your system must include a **retrievable knowledge layer**, not just prompts.

It should cover:

- Soil health (pH, organic carbon, moisture)
- Land use / land cover
- Biodiversity indicators (species richness, habitat diversity)
- Climate factors (temperature, rainfall)
- Human impact (pollution, deforestation)

Expectation:

- Use RAG, embeddings, vector databases, or structured datasets
- Index research papers, reports, or environmental datasets
- Clearly show how knowledge is retrieved and used

2. Conversational Intelligence

Your system should:

- Ask **clarifying questions** when inputs are incomplete
- Handle **multi-turn conversations with memory**
- Adapt responses based on context

Example:

User: "Biodiversity is declining on my land"

System: "Can you provide soil organic carbon %, rainfall pattern, and land use type?"

3. Evidence-Backed Recommendations (Mandatory)

Each recommendation must include:

- **What to do**
- **Why it works (scientific reasoning)**
- **Which environmental metric improves**
- **Reference to a study/report/model**

Expected Quality:

"Introduce legume-based cover crops → increases soil organic carbon by ~15–25% over 2–3 years (FAO studies), improving microbial diversity and pollinator support."

Not acceptable:

"Use sustainable practices"

4. Multi-Metric Reasoning

Your system must connect multiple variables:

- Soil health ↔ biodiversity
- Water availability ↔ species survival
- Land use ↔ habitat fragmentation

This is the core differentiator—**no single-variable answers**.

5. Input Handling

Support at least:

- Text input (mandatory)
- Structured input (JSON or similar)

Bonus:

- Geo-coordinates or spatial context

6. Output Quality

Each response must clearly include:

- Recommendation
- Impacted metrics
- Time horizon (short / medium / long term)
- Confidence level (optional but valuable)

Example Use Case

Input:

- Soil organic carbon: 0.3%
- Rainfall: low
- Crop: monoculture wheat
- Region: semi-arid

Expected Output:

- Suggest agroforestry / intercropping
- Explain impact on soil carbon and biodiversity
- Provide measurable improvement estimates
- Reference credible sources such as
 - Food and Agriculture Organization
 - Intergovernmental Panel on Climate Change

Evaluation Criteria

1. Depth of Reasoning (30%)

- Are recommendations non-obvious?
- Do they combine multiple environmental variables?

2. Scientific Grounding (25%)

- Are claims backed by credible sources?
- Is reasoning accurate and explainable?

3. Knowledge System Design (20%)

- Use of RAG / vector DB / structured datasets
- Clarity of knowledge retrieval pipeline

4. Conversational Intelligence (15%)

- Context awareness
- Follow-up questioning
- Memory handling

5. Output Clarity (10%)

- Structured, readable, and actionable responses

Constraints

- No generic LLM-only solutions
- No shallow or obvious recommendations

Constraints

- Must demonstrate knowledge grounding + reasoning.
- Must handle at least 3 environmental variables together.

Goal

Build a system that behaves like an AI environmental scientist, not a chatbot.

Documents Submission Guidelines

Submit one Word document (.docx) through the job where you applied, using the **My Jobs** page / **Applied Job** page and its document-submission option.

The Word document must include:

- 1 GitHub repository link for the completed project.
- 2 Live demo URL, where applicable.
- 3 A brief README.md overview covering the architecture, database/schema, local setup, and CI/CD details.
- 4 Any other links, credentials, or notes required to review and run the submission.

Repository access instructions

If the GitHub repository is private, grant access to the following accounts and include the repository link in the Word document:

- ankita.dasgupta@darukaa.com
- harsh.kumar@darukaa.com
- utkarsh.gauniyal@darukaa.com
- guneet.mutreja@darukaa.com

If the GitHub repository is public, provide the repository link only; access invitations are not required.

All submission details and links must be included in the Word document submitted through the applied-job page.